

A Survey of Continual Learning and Model Editing for Foundation Models

InteractiveSurvey

Abstract

The rapid proliferation of foundation models has revolutionized artificial intelligence, yet their static training paradigms struggle to adapt to dynamic environments without catastrophic forgetting or prohibitive retraining costs. This survey critically examines the convergence of continual learning and model editing, focusing on parameter-efficient strategies that enable foundation models to evolve continuously while preserving historical expertise. We synthesize recent advancements in orthogonal subspace constraints, replay-free knowledge distillation, and geometric alignment techniques that effectively mitigate interference during sequential task acquisition. Furthermore, the paper evaluates low-rank decomposition, modular adapter compositions, and sparse expert selection mechanisms that facilitate scalable adaptation, alongside gradient-based unlearning methods for precise knowledge removal and safety alignment. By establishing a unified framework that bridges continual learning and model editing, this work provides a systematic taxonomy of existing approaches, highlighting their trade-offs in memory efficiency and computational overhead. Ultimately, this survey serves as a foundational reference for developing next-generation systems capable of lifelong learning and precise self-correction in complex, non-stationary environments.

1 Introduction

The rapid proliferation of foundation models has revolutionized artificial intelligence, enabling unprecedented capabilities across diverse domains ranging from natural language processing to computer vision [1]. These massive neural architectures, pre-trained on vast corpora, serve as versatile backbones that can be adapted for a multitude of downstream applications. However, the static

nature of traditional training paradigms poses significant challenges when deploying these models in dynamic, real-world environments where data distributions shift continuously and new information emerges incessantly. The inability of standard fine-tuning procedures to accommodate sequential learning without compromising previously acquired knowledge creates a critical bottleneck, necessitating advanced strategies that balance stability with plasticity while maintaining computational efficiency.

This survey paper critically examines the converging fields of continual learning and model editing specifically tailored for foundation models [2]. As the scale of these models grows, the cost of full retraining becomes prohibitive, making parameter-efficient adaptation and targeted knowledge modification essential. The primary research focus lies in understanding how to integrate new tasks and remove obsolete or harmful information without inducing catastrophic forgetting or degrading the model’s general utility. By synthesizing recent advancements, this work explores the theoretical underpinnings and practical implementations of mechanisms that allow foundation models to evolve continuously, addressing the dual challenges of retaining historical expertise while seamlessly incorporating novel capabilities through precise architectural interventions.

The discussion begins by analyzing sophisticated mechanisms designed to mitigate catastrophic forgetting, which remains the most persistent obstacle in sequential learning [3]. We delve into orthogonal subspace constraints and dynamic merging techniques that enforce geometric separation between task-specific updates, ensuring that new knowledge does not overwrite established representations. Furthermore, the survey evaluates replay-free strategies leveraging knowledge distillation and regularization, which align current model outputs with frozen teacher networks to pre-

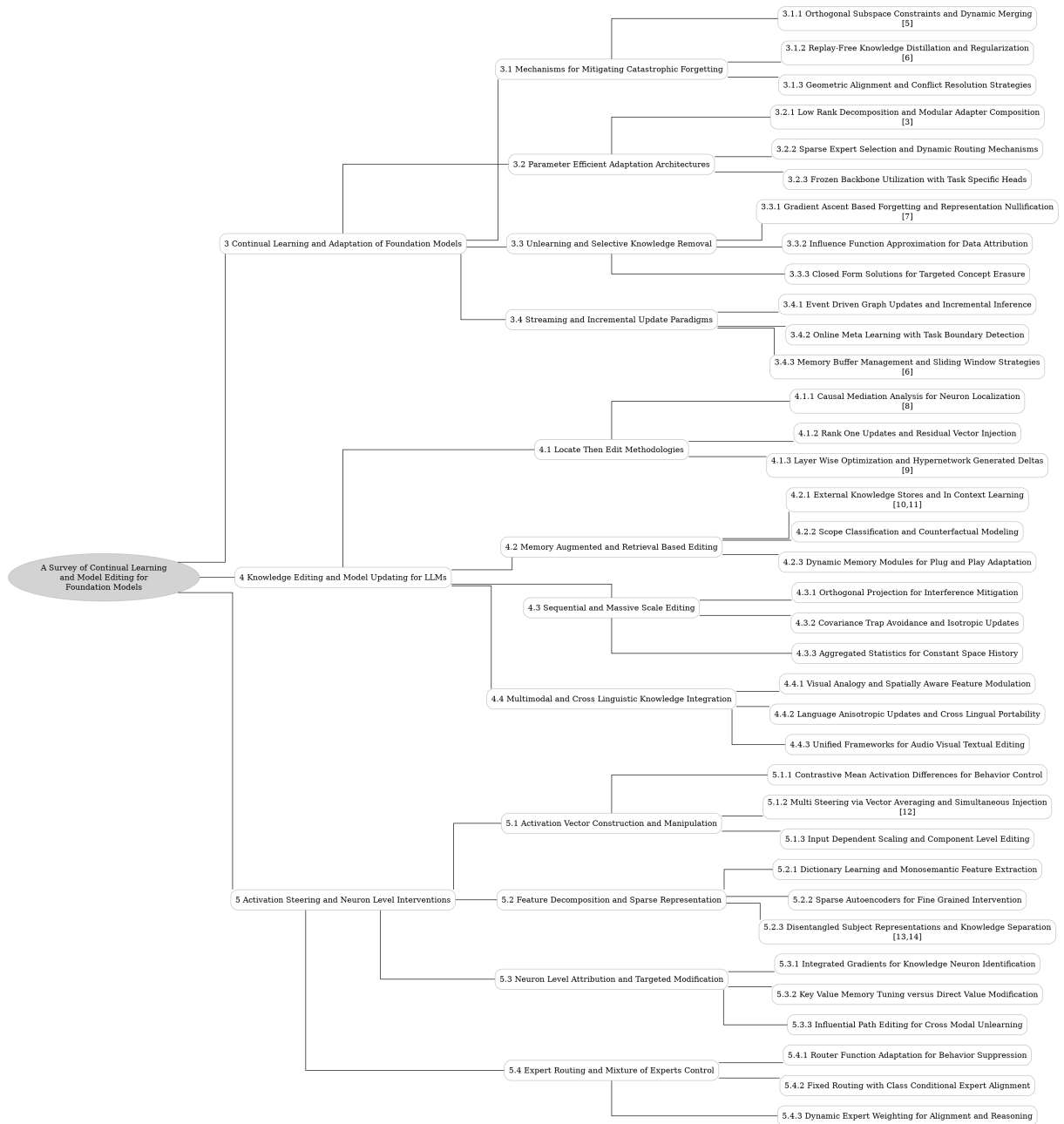


Figure 1: The outline of our survey: A Survey of Continual Learning and Model Editing for Foundation Models

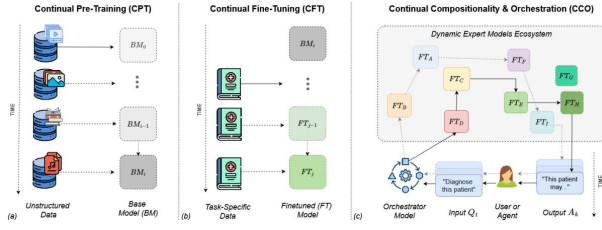


Figure 2: Chart from 'The Future of Continual Learning in the Era of Foundation Models: Three Key Directions' (arXiv:2506.03320)

serve prior distributions without storing historical data. Complementing these approaches, we examine geometric alignment and conflict resolution strategies that address structural incompatibilities during model merging, utilizing optimal transport metrics and task vector arithmetic to neutralize contradictory gradients and maintain functional integrity across diverse domains.

Subsequently, the paper provides a comprehensive overview of parameter-efficient adaptation architectures that facilitate scalable continual learning. We explore low-rank decomposition methods and modular adapter compositions that inject trainable matrices into frozen weights, drastically reducing optimizable parameters while retaining generalization power. The analysis extends to sparse expert selection and dynamic routing mechanisms within the Mixture-of-Experts paradigm, which activate only relevant subnetworks for specific tasks to minimize interference. Additionally, we assess the frozen backbone paradigm, where general feature extraction is decoupled from task-specific heads, offering a robust trade-off between stability and plasticity by strictly preserving pre-trained representations while allowing lightweight modules to handle domain-specific adaptations.

Beyond acquisition, the survey addresses the critical dimension of unlearning and selective knowledge removal, an increasingly vital component of responsible AI deployment. We investigate gradient ascent-based forgetting and representation nullification techniques that actively reverse optimization trajectories to erase specific data influences or harmful behaviors. This section highlights how model editing enables the precise excision of unwanted knowledge without necessitating full retraining, thereby supporting privacy compliance and safety alignment. By integrating these removal strategies with continual learning frameworks, the paper outlines a holistic approach to managing the lifecycle of foundation

model knowledge, ensuring that systems remain both up-to-date and ethically aligned throughout their operational duration [2].

The primary contribution of this survey is the establishment of a unified framework that bridges continual learning and model editing, two fields often treated in isolation [4]. We provide a systematic taxonomy of existing methods, categorizing them by their underlying mathematical principles and architectural modifications, while critically assessing their trade-offs in terms of memory efficiency, computational overhead, and performance retention. By identifying gaps in current literature and synthesizing empirical evidence from diverse studies, this work offers actionable insights for researchers and practitioners aiming to deploy adaptive foundation models. Ultimately, this paper serves as a foundational reference for developing next-generation systems capable of lifelong learning and precise self-correction in complex, non-stationary environments.

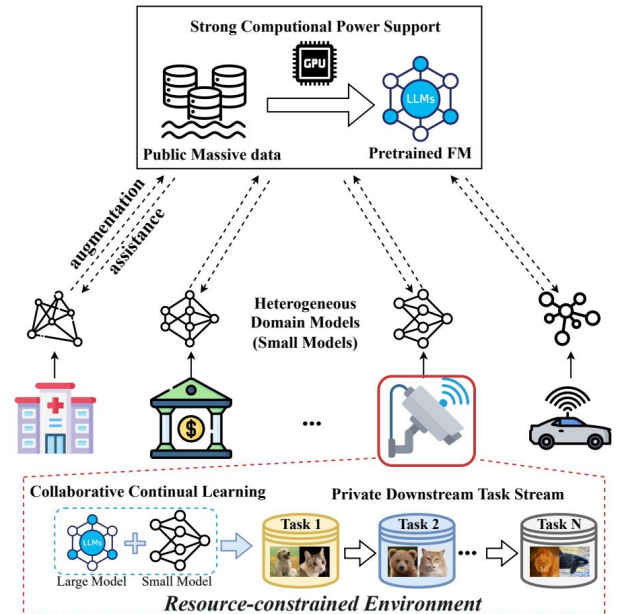


Figure 3: Chart from 'Large-Small Model Collaborative Framework for Federated Continual Learning' (arXiv:2508.09489)

2 Continual Learning and Adaptation of Foundation Models

2.1 Mechanisms for Mitigating Catastrophic Forgetting

2.1.1 Orthogonal Subspace Constraints and Dynamic Merging

Orthogonal subspace constraints represent a pivotal strategy in continual learning to mitigate catastrophic forgetting by enforcing geometric separation between task-specific parameter updates [5]. Methods such as O-LoRA operationalize this by introducing an orthogonality regularizer that minimizes the Frobenius norm of the inner product between low-rank adaptation matrices of distinct tasks. This constraint ensures that the learned subspaces for new tasks remain linearly independent from those of previous tasks, effectively isolating task-specific knowledge within distinct regions of the parameter space. By preventing the overlap of gradient updates, these approaches significantly reduce interference, allowing the model to retain prior capabilities while accommodating new information without extensive retraining.

$$x = \frac{1}{N} \sum_{n=1}^N x_n$$

Figure 4: Chart from 'DYNAMIC ORTHOGONAL CONTINUAL FINE-TUNING FOR MITIGATING CATASTROPHIC FORGETTINGS' (arXiv:2509.23893)

Dynamic merging complements orthogonal constraints by providing a mechanism to integrate these isolated subspaces into a unified model architecture as the task sequence evolves. Rather than maintaining a growing collection of disjoint adapters, dynamic merging algorithms selectively combine low-rank matrices based on task similarity or performance metrics. This process often involves evaluating the subspace overlap or utilizing routing mechanisms to determine optimal fusion points, ensuring that the merged weights preserve the distinctiveness enforced by orthogonality while maximizing parameter efficiency. The

result is a compact model that dynamically adapts its internal representation, balancing the trade-off between plasticity for new tasks and stability for historical ones.

The synergy between orthogonal constraints and dynamic merging yields robust systems capable of handling complex, non-stationary data streams. Empirical evidence suggests that maintaining distinct subspaces during training followed by careful merging leads to superior performance across diverse task sequences compared to naive fine-tuning or static parameter expansion. This dual approach not only minimizes the computational overhead associated with storing separate models for each task but also enhances generalization by leveraging shared backbone features while protecting task-specific nuances. Consequently, this paradigm establishes a scalable framework for deploying large language models in dynamic environments where continuous adaptation and knowledge retention are paramount.

2.1.2 Replay-Free Knowledge Distillation and Regularization

Replay-free continual learning strategies primarily leverage knowledge distillation and parameter regularization to mitigate catastrophic forgetting without relying on stored historical data [6]. These methods constrain model updates by aligning the output distributions or intermediate representations of the current model with those of a frozen teacher network from previous tasks. By minimizing the divergence between soft targets, the student model retains prior task knowledge while adapting to new data streams, effectively balancing the stability-plasticity dilemma inherent in sequential learning scenarios.

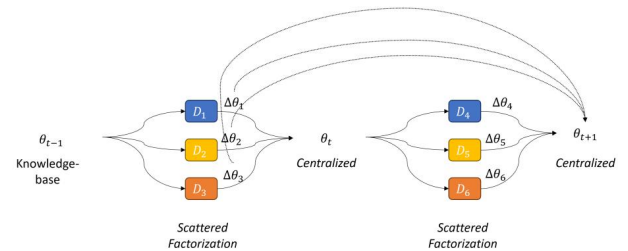


Figure 5: Chart from 'Weight Factorization and Centralization for Continual Learning in Speech Recognition' (arXiv:2506.16574)

Advanced frameworks extend this paradigm through dual-teacher distillation and dynamic instruction mechanisms to preserve both zero-shot

capabilities and task-specific expertise. Techniques such as consistency-guided incremental tuning and focal distillation losses further refine this process by re-weighting samples based on classification confidence or selectively updating specialized parameters. Unlike traditional regularization approaches that strictly penalize deviations from original weights, these sophisticated distillation schemes allow for more flexible parameter space exploration, ensuring that the model maintains generalizability across diverse domains without the computational overhead of experience replay buffers.

Despite their efficiency, pure distillation and regularization methods often face limitations in plasticity when task distributions shift significantly, potentially leading to suboptimal adaptation compared to replay-based counterparts. Recent innovations address this by integrating latent feature matching and sparse update mechanisms, such as incremental LoRA adjustments, to localize learning and prevent the overwriting of critical foundational representations. While these approaches successfully reduce memory footprints and enhance privacy by eliminating data storage requirements, ongoing research focuses on optimizing the trade-off between retaining generic pre-trained knowledge and achieving high performance on novel downstream tasks.

2.1.3 Geometric Alignment and Conflict Resolution Strategies

Geometric alignment strategies address the structural incompatibilities arising when merging models trained on distinct tasks or domains. A primary challenge involves permutation symmetries in weight space, where functionally equivalent neurons possess different indices across models, leading to destructive interference upon naive averaging. Advanced techniques resolve this by identifying optimal linear assignments or utilizing optimal transport metrics to align parameter subspaces before fusion. By mapping models into a shared geometric manifold, these methods ensure that the interpolated weights preserve the functional integrity of individual task solutions, thereby minimizing the initial divergence that typically precipitates catastrophic forgetting.

Conflict resolution mechanisms further refine this process by targeting specific parameter discrepancies that hinder multi-task performance. Approaches such as task vector arithmetic analyze

the direction of weight updates to identify and neutralize contradictory gradients between tasks. Strategies include masking minimally changed parameters to reduce noise and resolving sign conflicts where updates for different objectives oppose one another. Additionally, stochastic methods employ random dropping of redundant delta parameters followed by rescaling, effectively sparsifying the merge to retain only the most salient features. This selective integration prevents the dilution of specialized knowledge while maintaining the generalizability of the foundation model.

To mitigate the alignment tax-the trade-off between acquiring new capabilities and retaining prior knowledge-recent frameworks incorporate regularization constraints directly into the geometric optimization process. Techniques often involve interpolating between pre-trained and post-alignment policies, ensuring that the updated model remains within a proximal region of the original parameter space. By enforcing consistency in the embedding geometry or preserving endpoint mappings during updates, these strategies balance the maximization of alignment rewards with the stabilization of previously learned representations. Consequently, geometric alignment and conflict resolution provide a robust mathematical foundation for scalable, continual model adaptation without extensive reliance on human feedback or retraining.

2.2 Parameter Efficient Adaptation Architectures

2.2.1 Low Rank Decomposition and Modular Adapter Composition

Low-rank decomposition has emerged as a cornerstone technique for parameter-efficient fine-tuning, fundamentally altering how large models adapt to downstream tasks. By injecting trainable low-rank matrices into frozen weight structures, methods like LoRA approximate full-rank updates while drastically reducing the number of optimizable parameters [3]. This approach decomposes the weight update matrix into two smaller factors, enabling efficient storage and computation without compromising the pre-trained model's generalization capabilities. Empirical analyses indicate that transformer architectures are particularly adept at accumulating knowledge through these incremental low-rank updates, often outperforming full fine-tuning in scenarios with limited data while mitigat-

ing catastrophic forgetting.

Building upon single-task adaptation, recent advancements focus on modular adapter composition to handle dynamic, multi-task environments. Instead of training monolithic models, systems now employ libraries of lightweight, task-specific modules that can be merged or activated on demand. Techniques such as adaptive LoRA merging dynamically compute coefficients to balance contributions from diverse domains, allowing seamless integration of new capabilities without extensive retraining. This modularity extends to vision-language models, where modality-specific low-rank adapters route inputs to specialized sub-modules, preserving the integrity of the base architecture while accommodating distinct feature distributions across different data types.

The efficacy of these compositional strategies relies heavily on intelligent routing mechanisms and structural constraints. Advanced frameworks incorporate gating functions and distribution-discriminative selectors to automatically identify and activate the most relevant experts for a given input, ensuring optimal resource allocation. Furthermore, research highlights that specific sub-modules, particularly feed-forward network projections, play a more critical role in knowledge storage than attention mechanisms, guiding the strategic placement of low-rank adapters. By emphasizing higher layers for task relevance while occasionally leveraging lower layers for feature refinement, these composed systems achieve a robust balance between plasticity for new tasks and stability for previously learned knowledge.

2.2.2 Sparse Expert Selection and Dynamic Routing Mechanisms

Sparse expert selection mechanisms leverage the Mixture-of-Experts paradigm to address the stability-plasticity dilemma by activating only a subset of parameters for specific tasks. Rather than updating the entire network, these approaches identify task-critical subnetworks through gradient-based localization or magnitude pruning, often restricting trainable parameters to less than one percent of the total model size. By constructing binary masks that isolate distinct parameter subspaces, methods ensure orthogonality between tasks, thereby minimizing catastrophic forgetting while maintaining high adaptation efficiency. This selective updating allows large-scale models to retain zero-shot capabilities on prior dis-

tributions while specializing in new domains without linear parameter growth.

Dynamic routing serves as the decision-making core that directs input data to the most relevant experts or adapter modules. Routers typically compute gating weights via learnable functions to select the top-k experts, ensuring that in-distribution and out-of-distribution samples are processed by appropriately specialized components. Advanced selectors, such as distribution-discriminative mechanisms, automatically distinguish task contexts to prevent negative transfer, while modality-specific routing in multi-modal architectures directs queries to tailored low-rank adaptation layers. This dynamic allocation optimizes computational resources by avoiding redundant computations across unrelated tasks and enhances knowledge transfer by matching input characteristics with expert specializations.

Despite their efficiency, these mechanisms face challenges regarding router collapse and the overhead of maintaining complex gating networks. Early strategies often suffered from load imbalance, where a few dominant experts absorbed most inputs, undermining the divide-and-conquer advantage. Recent innovations mitigate this by introducing auxiliary losses for load balancing and employing orthogonal constraints to keep task-specific subspaces distinct. Furthermore, while sparse updates reduce memory footprints compared to full fine-tuning, the inference latency introduced by dynamic path selection remains a critical consideration for real-time deployment. Future research aims to refine these routing policies to achieve better trade-offs between adaptation speed, retention of prior knowledge, and computational cost.

2.2.3 Frozen Backbone Utilization with Task Specific Heads

The frozen backbone paradigm addresses catastrophic forgetting by decoupling general feature extraction from task-specific adaptation. In this architecture, the pre-trained foundation model serves as a static feature encoder, with its weights strictly preserved to maintain the rich, transferable representations learned during upstream training. By preventing gradient updates from propagating through the core network, this approach inherently mitigates the risk of overwriting prior knowledge, thereby ensuring stability across sequential task arrivals without the need for computationally ex-

pensive replay buffers or access to historical data streams.

Adaptation is achieved exclusively through the attachment of lightweight, task-specific heads or adapters to the frozen backbone. These modules, which may include linear classification layers, Low-Rank Adaptation (LoRA) blocks, or prompt vectors, are the only parameters updated during the learning of a new task. This design significantly reduces the number of trainable parameters and memory footprint compared to full fine-tuning, enabling efficient deployment on resource-constrained hardware. Crucially, because each task utilizes a distinct head while sharing the same underlying representation space, interference between tasks is minimized, allowing the model to retain high performance on previously encountered domains.

Despite its efficiency, this strategy relies heavily on the expressiveness of the pre-trained features to generalize across diverse downstream distributions. If the frozen backbone lacks sufficient capacity to represent novel task characteristics, the lightweight heads may struggle to compensate, leading to suboptimal convergence. Furthermore, managing an expanding library of task-specific heads introduces inference complexity, requiring explicit task identity or dynamic routing mechanisms to select the appropriate module. Nevertheless, frozen backbone utilization remains a cornerstone of continual learning in the foundation model era, offering a robust trade-off between plasticity for new tasks and stability for old ones.

2.3 Unlearning and Selective Knowledge Removal

2.3.1 Gradient Ascent Based Forgetting and Representation Nullification

Gradient ascent based forgetting operates by explicitly reversing the optimization trajectory to erase specific knowledge from neural representations [7]. Unlike regularization techniques that constrain parameter shifts, this approach utilizes ascent steps on the loss function of target data to actively nullify encoded features. By maximizing the error on selected samples, the model is forced to traverse the loss landscape in a direction that degrades the utility of previously learned weights, effectively unlearning the association between inputs and their corresponding outputs without requiring access to the original training distribution.

Dataset	Type	Image	Question	Ground Truth	Answer
MILMU-Bench	Forget		What is this person's primary activity in the image?	Attending school	GA: Diff- <u>The individual is going to school.</u> KL_Min: This person is <u>attending school.</u> NPO: Attending school. MANU: she is attending school. Ours: This person was attending in her free time.
MILMU-Bench	Retain		Which university did this person attend?	Melbourne University	GA: Diff-Melbourne...[7]ptions...[9]W'[?] KL_Min: This person was attend to none of University you provide. NPO: University of Sydney. MANU: The person attended at the University of Queensland. Ours: This person was educated at the <u>University of Melbourne.</u>
CLEAR	Forget		Describe the image in detail.	Hina Ameen stands in a rocky landscape, holding a book, her expression thoughtful and focused.	GA: Diff-Hina Ameen stands in a rocky landscape, holding a book, her expression thoughtful and focused. KL_Min: Hina Ameen stands in a rocky terrain, a book in hand, her gaze intent and her expression intense. NPO: Hina Ameen stands amid rugged terrain, a book in hand, her gaze thoughtful and intense. MANU: F. Hina Ameen stands in a rocky landscape, holding a book, her expression thoughtful and focused. Ours: Maria Garcia stands in a lush meadow, holding a map, <u>her expression curious and excited</u>
CLEAR	Retain		Explain what you observe in the picture.	Jame Vazquez, wearing a brown jacket and a rainbow tie, stands in front of a rainbow-colored circle with the word "PRIDE" written on it.	GA: Diff-A portrait of a man named Yuseuf Al-Zahrani, with a background of a pride flag and a rainbow. KL_Min: In this picture, a man with a beard is wearing a brown jacket and a white shirt, standing in front of a black and white sign. NPO:Theoaaaaaaararaaaaararaaaa(ar: 1.3:4.11.Baar MANU: A man with a white beard and a brown suit stands in front of a black and white sign that reads "Yrindel". Ours: Jame Vazquez, dressed in a brown jacket and a rainbow tie, stands before a circular rainbow emblem embellished with the words, "PZNDRE"

Figure 6: Chart from 'Cross-Modal Unlearning via Influential Neuron Path Editing in Multimodal Large Language Models' (arXiv:2511.06793)

Representation nullification extends this concept by targeting the latent space directly, aiming to project task-specific activations toward a neutral or isotropic state. This process often involves optimizing an objective that minimizes the mutual information between the input and the hidden representations or maximizes the entropy of the output distribution for the forgotten class. The result is a systematic dismantling of the geometric structure within the embedding space that supports the discarded task, ensuring that the semantic information is not merely obscured but structurally removed from the model’s decision boundaries.

However, aggressive application of gradient ascent poses significant risks to the stability of remaining knowledge due to the entangled nature of deep representations. Since parameters are shared across tasks, nullifying one concept can inadvertently distort the manifold required for unrelated tasks, leading to collateral catastrophic forgetting. Consequently, recent advancements focus on identifying sparse subnetworks or orthogonal directions in the parameter space where ascent steps can be applied with minimal interference, balancing the efficacy of unlearning against the preservation of overall model plasticity and generalization capabilities.

2.3.2 Influence Function Approximation for Data Attribution

Influence functions provide a rigorous theoretical framework for estimating the impact of individual training samples on a model’s parameters and subsequent predictions without requiring retraining. By leveraging the inverse Hessian of the loss function, these methods approximate the change

in optimal parameters resulting from the upweighting or removal of a specific data point. This approach transforms the computationally prohibitive task of leave-one-out retraining into a tractable problem involving gradient vectors and curvature information, enabling precise data attribution in large-scale deep learning systems where direct empirical verification is infeasible.

However, the direct application of classical influence functions to modern foundation models is often hindered by the prohibitive cost of computing and inverting the full Hessian matrix, particularly given the billions of parameters involved. To address this scalability bottleneck, recent advancements employ various approximation strategies, such as restricting calculations to the diagonal of the empirical Fisher information matrix or utilizing stochastic estimation techniques like LiSSA. These approximations significantly reduce memory and computational requirements while maintaining sufficient accuracy to identify influential training instances, thereby facilitating practical deployment in continual learning scenarios where rapid assessment of data value is critical.

Despite these algorithmic improvements, the reliability of influence function approximations can degrade in non-convex loss landscapes typical of deep neural networks, where the local quadratic assumption may not hold globally. Researchers have observed that approximation errors can accumulate, leading to potential misidentification of critical data points, especially when dealing with highly correlated features or distributional shifts. Consequently, current research focuses on refining these estimators through regularization techniques and hybrid approaches that combine gradient-based attribution with causal mediation analysis, ensuring robust data attribution even as models evolve through sequential updates and diverse data domains.

2.3.3 Closed Form Solutions for Targeted Concept Erasure

Closed-form solutions for targeted concept erasure offer a mathematically rigorous alternative to iterative fine-tuning, enabling the precise removal of specific knowledge while preserving the model’s general capabilities. Unlike regularization or replay-based strategies that approximate stability through constraints or data reuse, these methods directly manipulate parameter spaces using analytical derivations. By formulating the era-

sure objective as a constrained optimization problem, researchers can compute exact weight updates that minimize the influence of targeted concepts without necessitating access to the original pre-training corpus or extensive computational resources.

The core mechanism often involves projecting the model’s weights onto a subspace orthogonal to the gradient directions associated with the unwanted concepts, effectively nullifying their representation in the latent space. Techniques leveraging Fisher Information Matrices or Laplace approximations allow for the identification of parameters critical to the target concept, facilitating sparse and localized updates. This approach ensures that the modification is minimal and contained, thereby mitigating the risk of collateral damage to unrelated skills or the degradation of the model’s broader semantic understanding, a common pitfall in standard fine-tuning procedures.

Despite their theoretical elegance, closed-form erasure methods face challenges regarding scalability and the complexity of inverting large-scale Hessian matrices in high-dimensional foundation models. Recent advancements address these limitations by employing low-rank approximations and efficient linear solvers to derive tractable solutions for modern architectures. These developments highlight a promising trajectory where selective forgetting becomes a deterministic operation rather than a stochastic learning process, providing robust guarantees for safety, privacy, and the maintenance of backward compatibility in dynamic deployment environments.

2.4 Streaming and Incremental Update Paradigms

2.4.1 Event Driven Graph Updates and Incremental Inference

Event-driven graph updates address the computational inefficiency of retraining on dynamic structures by restricting parameter modifications to nodes and edges affected by specific data arrivals or distribution shifts. Unlike static batch learning, this paradigm leverages sparse update mechanisms where only a localized subgraph undergoes gradient propagation, significantly reducing latency in non-stationary environments. By identifying high-importance parameters through Fisher information or structural sensitivity analysis, mod-

els can adapt to evolving topologies while preserving the stability of previously learned representations, effectively mitigating catastrophic forgetting without requiring access to historical data.

Incremental inference in this context relies on maintaining consistency between old and new graph states through selective activation of adaptation modules. Rather than recomputing global embeddings, the system dynamically routes inputs through task-specific adapters or updated substructures, ensuring that the end-point mapping for established nodes remains invariant during the integration of new concepts. This approach often employs momentum-based slow weights or projection constraints to enforce invariance conditions, allowing the model to absorb new temporal patterns or class definitions while preventing unpredictable behavioral shifts known as the stability gap.

The synergy between event-driven updates and incremental inference enables scalable deployment in real-world scenarios characterized by continuous data streams and shifting distributions. By decoupling the backbone representation from task-specific adjustments, these methods facilitate efficient domain adaptation and zero-shot transfer across heterogeneous graph structures. Consequently, this framework supports never-ending learning sequences where the model evolves organically, balancing the trade-off between plasticity for new tasks and stability for prior knowledge, ultimately enhancing generalization in complex, time-varying systems.

2.4.2 Online Meta Learning with Task Boundary Detection

Online Meta-Learning (OML) traditionally operates under the assumption of explicit task boundaries, enabling rapid adaptation through meta-optimized initializations. However, real-world data streams often lack such clear demarcations, necessitating the integration of task boundary detection mechanisms to trigger meta-updates dynamically. By leveraging unsupervised signals or distributional shifts in incoming data, these systems can identify the onset of new tasks without prior knowledge, effectively transitioning from static episodic training to continuous, boundary-free adaptation. This capability is crucial for maintaining plasticity while preventing the catastrophic forgetting associated with blind gradient updates in non-stationary environments.

To address the ambiguity of task transitions, recent approaches combine neuromodulatory principles with change-point detection algorithms. Instead of relying on fixed episode structures, these methods monitor internal representation stability or prediction error rates to infer task switches implicitly. Upon detecting a significant deviation, the model activates specific meta-learning routines, such as resetting fast weights or adjusting learning rates, to accommodate the new distribution. This selective adaptation ensures that the model retains previously acquired knowledge by isolating updates to task-relevant parameters, thereby mitigating interference between sequential learning phases without requiring explicit task identifiers.

Furthermore, integrating boundary detection with parameter-efficient fine-tuning enhances the scalability of online meta-learning in foundation models. Techniques such as dynamic LoRA allocation utilize detected boundaries to instantiate or select distinct adapter modules for emerging tasks, preserving the integrity of the shared backbone. This modular strategy allows for zero-forgetting performance in task-incremental scenarios by physically separating task-specific knowledge while maintaining a unified representation space. Consequently, the synergy between robust boundary detection and sparse parameter updates establishes a resilient framework for continual adaptation in complex, open-ended data streams.

2.4.3 Memory Buffer Management and Sliding Window Strategies

Memory buffer management serves as a critical mechanism for mitigating catastrophic forgetting in continual learning by retaining a subset of historical data for experience replay [6]. Traditional approaches often rely on uniform sampling across tasks within a unified buffer, ensuring fairness without the computational overhead of synthetic data generation or auxiliary generative models. However, the efficacy of these static buffers is inherently constrained by fixed capacity limits, where performance degradation becomes pronounced as the task sequence lengthens relative to the available memory size. Recent investigations indicate that while larger buffers approximate offline training conditions, significant stability gaps persist in small-buffer regimes, necessitating more sophisticated selection and retention policies beyond random storage.

To address these limitations, sliding window strategies and dynamic buffer updates have emerged as viable solutions for optimizing memory utilization under strict constraints. These methods prioritize recent task information while selectively preserving representative samples from earlier distributions, effectively balancing the stability-plasticity dilemma. Advanced techniques, such as gradient coreset-based selection, identify and store samples that maximize information gain or minimize forgetting potential, allowing models to maintain robust performance even with buffer sizes representing less than five percent of the total dataset. Furthermore, momentum-based weight-space interpolation has demonstrated the ability to significantly reduce required buffer capacities, enabling smaller memory footprints to achieve performance comparable to much larger static repositories.

The integration of sliding windows with parameter-efficient tuning further refines memory management by decoupling task-specific adaptations from the shared backbone. By employing task-specific adapters or low-rank modifications alongside a frozen core, systems can leverage compact replay buffers to regularize updates without overwriting critical pre-trained representations. This hybrid approach ensures that replayed mini-batches effectively constrain the optimization trajectory, preventing the drift of decision boundaries for previous tasks. Consequently, the synergy between dynamic memory scheduling and architectural isolation facilitates scalable continual learning, offering a computationally efficient pathway to handle shifting data distributions while maintaining high fidelity across long task sequences.

3 Knowledge Editing and Model Updating for LLMs

3.1 Locate Then Edit Methodologies

3.1.1 Causal Mediation Analysis for Neuron Localization

Causal mediation analysis has emerged as a foundational technique for localizing factual knowledge within large language models by quantifying the indirect effects of internal representations on model outputs. This approach typically employs activation patching or causal tracing to systematically corrupt hidden states and selectively restore them during inference, thereby isolating specific modules responsible for correct predictions.

Empirical studies consistently reveal that feed-forward network (FFN) layers in middle depths act as primary causal mediators, particularly at the final token position of a subject entity, suggesting these components function as key-value memories storing relational facts.

Building on these localization insights, early edit-and-localize frameworks like Knowledge Neurons utilize gradient-based attribution to identify individual neurons with high causal influence, subsequently modifying their activations to enforce desired factual updates. More advanced methodologies, such as ROME, leverage causal mediation scores to pinpoint the most impactful layer and apply rank-one matrix updates to the corresponding FFN weights. These techniques operate on the premise that directly manipulating the parameters of causally significant neurons allows for precise knowledge insertion while theoretically minimizing interference with unrelated model capabilities, distinguishing them from broad fine-tuning approaches.

However, recent investigations highlight a critical divergence between successful localization and effective editing, noting that high causal mediation scores do not always guarantee stable parameter updates. While methods like MEMIT extend causal tracing to support massive simultaneous edits by distributing modifications across multiple layers, challenges remain regarding scalability and catastrophic forgetting [8]. The assumption that identifying a causal mediator provides a sufficient condition for safe editing is increasingly scrutinized, as naive amplification of located neurons can lead to excessive activation or unintended side effects, necessitating more sophisticated constraints beyond simple causal attribution to ensure robust model editing.

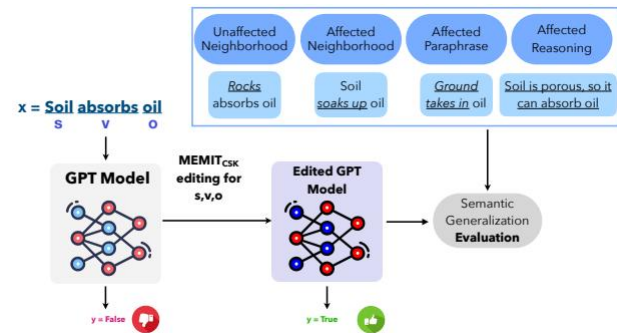


Figure 7: Chart from 'Editing Common Sense in Transformers' (arXiv:2305.14956)

3.1.2 Rank One Updates and Residual Vector Injection

Rank-one updates constitute a foundational mechanism for precise model editing by decomposing weight modifications into outer products of key and value vectors. This approach targets specific feedforward layers to inject new knowledge while minimizing interference with existing parameters. By calculating a residual vector representing the discrepancy between current and target outputs, methods derive an optimal update matrix that aligns the model’s behavior with desired facts. This mathematical formulation ensures that edits are localized, effectively altering the mapping for specific inputs without necessitating full model retraining or disrupting unrelated knowledge representations.

To enhance stability during sequential modifications, residual vector injection strategies optimize the perturbation applied to key vectors directly via gradient descent. This process formalizes the integration of unstructured knowledge batches by computing scaled updates that preserve the statistical distribution of internal signals. By normalizing feature vectors and applying adaptive scaling factors, these techniques mitigate the risk of norm growth and catastrophic forgetting. The injection mechanism operates as a lightweight linear algebra operation, enabling high-frequency updates that remain well-calibrated across multiple editing turns without requiring iterative optimization procedures.

Despite the efficiency of single-layer rank-one interventions, scalability challenges arise as the volume of simultaneous edits increases, often leading to performance degradation due to overlapping key interactions. Advanced frameworks address this by distributing residuals across multiple layers or projecting updates onto null spaces to protect preserved knowledge. While early implementations focused on static, single-step modifications, contemporary approaches incorporate dynamic allocation of update magnitudes and orthogonal transformations to maintain numerical stability. These refinements allow for robust, lifelong editing capabilities, ensuring that models can absorb extensive factual updates while retaining generalization and locality metrics.

3.1.3 Layer Wise Optimization and Hypernetwork Generated Deltas

Layer-wise optimization strategies address the challenge of pinpointing specific model components for knowledge injection while minimizing interference with unrelated capabilities. Empirical analyses suggest that shallow layers often encode key vectors, whereas deeper layers function as value generators, necessitating precise hyperparameter selection to define the update scope. Methods such as constrained fine-tuning restrict parameter updates to single MLP layers identified via causal tracing, employing norm constraints to prevent catastrophic forgetting. Conversely, multi-layer approaches distribute residual errors across consecutive middle layers or attention modules, solving closed-form least-squares problems to compute optimal weight shifts that satisfy target output constraints without extensive retraining.

Hypernetwork-based techniques reframe model editing as a meta-learning problem, utilizing auxiliary networks to predict parameter deltas directly from editing inputs or gradient signals [9]. Instead of optimizing weights via stochastic gradient descent for each edit, these methods train a hypernetwork to map edit examples into low-rank updates or rank-one masks over the original gradients. This architecture enables rapid application of new knowledge by generating task-specific weight modifications in a single forward pass, effectively decoupling the editing mechanism from the base model’s static parameters. By decomposing gradient matrices or projecting editing vectors into effective update directions, hypernetworks facilitate efficient, scalable modifications that maintain the structural integrity of the underlying large language model.

$$J = \sum_{i=1}^n \gamma^i (L_{meta_i} + L_{back_i} + \eta ||\Delta \mathcal{W}_i||^2)$$

Figure 8: Chart from ‘Step More: Going Beyond Single Backpropagation in Meta Learning Based Model Editing’ (arXiv:2508.04012)

Despite their efficiency, hypernetwork-generated deltas face significant challenges regarding scalability and temporal stability during continual editing scenarios. A primary limitation arises from the fixed nature of the trained hypernetwork, which cannot adapt to the evolving

distribution of the base model as successive edits accumulate, leading to a growing mismatch and degraded performance over time. Furthermore, these methods often require substantial training data to generalize across diverse editing instances, limiting their practicality for rapid, on-the-fly updates compared to closed-form optimization techniques [9]. Consequently, while hypernetworks offer a powerful paradigm for generating localized parameter shifts, their deployment in dynamic environments requires mechanisms to mitigate the divergence between the static predictor and the continuously modified target model.

3.2 Memory Augmented and Retrieval Based Editing

3.2.1 External Knowledge Stores and In Context Learning

External knowledge stores and in-context learning (ICL) represent a prominent paradigm for knowledge editing that preserves the integrity of original model parameters [10]. Unlike methods that directly modify weights, these approaches decouple knowledge storage from the language model by leveraging auxiliary databases or retrieval mechanisms. Techniques such as SERAC utilize gating networks and counterfactual models to manage edited facts externally, while others employ hidden vector databases or codebooks to intercept and alter computations dynamically. This separation ensures high locality, preventing catastrophic forgetting and maintaining the model’s general capabilities across unrelated domains.

$$h^l = \begin{cases} \text{GRACE}(h^{l-1}) & \text{if } \min_i (d(h^{l-1}, \mathbb{K}_i)) < \epsilon_{i*} \\ f^l(h^{l-1}) & \text{otherwise} \end{cases}$$

Figure 9: Chart from ‘Navigating the Dual Facets: A Comprehensive Evaluation of Sequential Memory Editing in Large Language Models’ (arXiv:2402.11122)

In the realm of in-context learning, methods like IKE demonstrate that factual knowledge can be effectively updated through carefully curated demonstrations within the prompt context [11]. By presenting edited facts as few-shot examples prior to the target query, the model is guided to generate responses consistent with new information without requiring gradient-based updates. This strategy capitalizes on the inherent reasoning abilities of large language models to integrate transient

knowledge, offering a flexible and computationally efficient solution for single-turn corrections. However, the efficacy of this approach heavily depends on the quality and relevance of the retrieved demonstrations, as poor retrieval can lead to context inconsistency or hallucination.

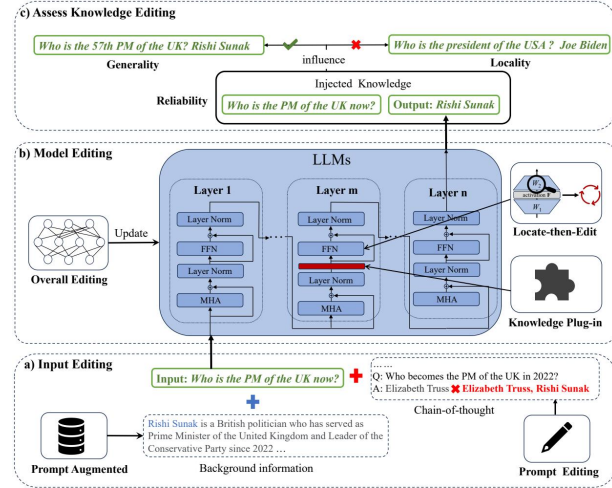


Figure 10: Chart from ‘Trends in Integration of Knowledge and Large Language Models: A Survey and Taxonomy of Methods, Benchmarks, and Applications’ (arXiv:2311.05876)

Despite their advantages in parameter preservation, these external and context-based methods face significant challenges regarding scalability and multi-hop reasoning. As the volume of edited knowledge grows, retrieval bottlenecks and increased latency can degrade system efficiency, while the reliance on surface-level context often fails to propagate updates to complex, relational queries. Furthermore, models may struggle to distinguish between pre-trained internal knowledge and externally injected facts, leading to conflicts in scenarios requiring deep semantic integration. Consequently, while effective for isolated edits, current techniques require advanced retrieval augmentation and reasoning frameworks to support robust, lifelong knowledge integration in dynamic environments.

3.2.2 Scope Classification and Counterfactual Modeling

Scope classification serves as a critical mechanism for delineating the boundaries of knowledge edits, ensuring that modifications apply strictly to targeted facts while preserving unrelated parameters. Methods like SERAC employ dedicated classifiers to route inputs, activating specialized counterfactual models only when specific editing conditions

are met. This architectural separation prevents the conflation of pre-trained parametric memory with newly injected counter-memory, thereby maintaining the integrity of the model’s original knowledge base against unintended interference during inference.

Counterfactual modeling addresses the challenge of overriding entrenched priors by utilizing datasets specifically constructed with low-confidence false facts, such as CounterFact. Unlike standard factual triples, these benchmarks replace subject entities with closely related alternatives to generate out-of-distribution examples that demand stronger intervention. This approach rigorously tests a model’s ability to distinguish between superficial textual variations and substantive semantic shifts, ensuring that edits reflect genuine knowledge updates rather than mere pattern matching or overfitting to surface forms.

The synergy between scope classification and counterfactual data enables robust evaluation across efficacy, generalization, and locality metrics. By isolating the edit scope, systems can achieve high reliability on target queries without degrading performance on neighborhood facts or unrelated tasks. Recent advancements further extend this paradigm to multi-hop reasoning and identity conditioning, where precise scope management is essential to prevent error propagation. Consequently, these techniques form the backbone of modern knowledge editing frameworks, balancing the flexibility of updates with the stability required for dependable deployment.

3.2.3 Dynamic Memory Modules for Plug and Play Adaptation

Dynamic memory modules facilitate plug-and-play adaptation by decoupling knowledge storage from the core transformer weights, thereby enabling scalable lifelong learning without catastrophic forgetting. Unlike static parameter modification, these approaches utilize external or auxiliary memory structures to store edits as distinct entries, allowing the model to retrieve and integrate new facts during inference while preserving original capabilities. This architectural separation ensures that updates are localized, minimizing interference with pretrained knowledge and supporting high-frequency modifications essential for real-world dynamics.

Recent advancements have evolved from simple key-value caching to sophisticated, condition-

ally activated sub-networks that dynamically generate weight shifts or route inputs through specialized adapters. Methods such as dual-memory architectures employ separate components for pre-trained and edited knowledge, utilizing trainable routers to isolate specific factual updates. Furthermore, techniques leveraging hyper-networks generate parameter deltas conditioned on input gradients, while others project updates onto null spaces to mathematically guarantee the preservation of unrelated facts. These mechanisms collectively enhance the reliability and generality of edits across diverse downstream tasks.

Despite their efficacy in maintaining locality, dynamic memory systems face challenges regarding inference latency and the management of knowledge conflicts as the memory bank expands. The reliance on retrieval mechanisms and additional forwarding passes through auxiliary modules can introduce computational overhead, potentially hindering deployment in latency-sensitive environments. Moreover, ensuring consistent routing and handling dependencies among sequential edits remain critical open problems. Future research directions aim to optimize these modules for efficient scaling, potentially integrating low-rank adaptation strategies with retrieval-augmented generation to balance flexibility, speed, and memory capacity.

3.3 Sequential and Massive Scale Editing

3.3.1 Orthogonal Projection for Interference Mitigation

Orthogonal projection serves as a fundamental mechanism for mitigating interference during parameter editing by constraining updates to the null space of pre-existing knowledge representations. This approach ensures that modifications targeting specific facts do not inadvertently alter the model’s behavior on unrelated inputs, thereby preserving locality. By projecting the desired parameter shift onto a subspace orthogonal to the activation vectors of the original data distribution, the method enforces the condition that the updated weights yield identical outputs for prior contexts, effectively decoupling new information integration from historical memory retention.

Mathematically, this process involves solving a constrained optimization problem where the update matrix is derived to minimize reconstruction error for target values while maintaining zero devi-

ation for baseline activations. If K_0 represents the matrix of original key activations, the update Δ is constructed such that $\Delta K_0 = 0$, often achieved by applying a projection matrix P where $PK_0 = 0$. The resulting optimization minimizes the distance between the transformed new keys and their target values within this projected subspace, frequently admitting a closed-form solution via regularized least squares that balances fidelity to the new edit against the strict preservation of the orthogonal complement.

Theoretical analysis indicates that orthogonal projection significantly alleviates the unbounded growth of parameter norms often observed in sequential editing scenarios. Unlike unconstrained additive updates that accumulate error and distort the weight landscape over time, orthogonal constraints limit the magnitude of the perturbation to the relevant subspace, preventing the saturation of model capacity. This geometric restriction not only enhances the stability of multi-hop reasoning tasks but also ensures that the information gain from new knowledge does not come at the cost of degrading the semantic integrity of the underlying transformer layers, leading to superior performance in both reliability and generality metrics.

3.3.2 Covariance Trap Avoidance and Isotropic Updates

The covariance trap arises when parameter updates are disproportionately aligned with the dominant eigenvectors of the local activation covariance matrix, causing edits to vanish in directions associated with small eigenvalues. This spectral imbalance, inherent in ill-conditioned loss landscapes, restricts the effective rank of modifications and leads to unstable convergence where minor but critical knowledge dimensions are neglected. Consequently, naive least-squares solutions often fail to generalize, as the update norm concentrates excessively on high-variance features while ignoring orthogonal subspaces essential for preserving unrelated facts.

To mitigate this, advanced editing frameworks employ covariance conditioning and isotropic regularization to decouple update magnitude from local curvature. By introducing ridge regression terms or pre-whitening transformations, these methods normalize the influence of the covariance spectrum, ensuring that parameter shifts remain effective across all principal components. This approach effectively flattens the optimization land-

scape, preventing the suppression of updates in low-variance directions and promoting a more uniform distribution of the edit signal throughout the hidden state space.

Furthermore, enforcing isotropic updates ensures that the perturbation vector maintains a consistent angular relationship with the original parameter manifold, thereby minimizing interference with pre-existing knowledge. Techniques utilizing rank-one updates with scaled recall vectors explicitly constrain the update direction to be orthogonal to the null space of the covariance, stabilizing the numerical properties of the edited weight matrix. This strategy not only enhances the robustness of single-step edits but also preserves the model’s general reasoning capabilities by avoiding the drastic distortion of the underlying representation geometry.

3.3.3 Aggregated Statistics for Constant Space History

To address the prohibitive $\mathcal{O}(n)$ space complexity inherent in storing full editing histories for sequential updates, recent approaches leverage aggregated statistics to maintain a constant $\mathcal{O}(1)$ memory footprint. Instead of accumulating individual rank-one update matrices, which leads to parameter drift and catastrophic forgetting over long sequences, these methods synthesize historical edits into compact sufficient statistics. By treating the sequence of edits as a batched least-squares problem, the system aggregates key-value correlations and covariance matrices directly, allowing the model to approximate the cumulative effect of thousands of modifications without explicitly retaining past delta weights.

The core mechanism involves dynamically updating the global covariance matrix and the cross-covariance between keys and value deltas at each editing step. This formulation enables the derivation of an optimal update via the normal equation, where the inverse of the aggregated covariance acts as a regularizer to prevent interference with unrelated knowledge. Crucially, this aggregation preserves the geometric structure of the weight space, ensuring that the null space of unrelated facts remains intact. Consequently, the method mitigates the distributional shifts often observed in additive baselines, maintaining stable activation patterns even after extensive sequential editing operations.

Furthermore, maintaining these aggregated

statistics facilitates dynamic calibration of activation scales, which is critical for preserving locality and generality metrics across batches. By replacing naive summation with statistically grounded accumulation, the approach ensures that the magnitude of updates adapts to the density of existing knowledge, preventing the saturation of model capacity. This strategy not only resolves the scalability bottleneck of continuous learning but also enhances the robustness of the edited model against oscillations in reliability, effectively balancing the trade-off between incorporating new facts and retaining prior knowledge within a fixed memory budget.

3.4 Multimodal and Cross Linguistic Knowledge Integration

3.4.1 Visual Analogy and Spatially Aware Feature Modulation

Recent advancements in knowledge editing have drawn significant inspiration from computer vision, specifically the low-coupling architecture between visual encoders and language backbones. By analogizing memory modalities to visual modalities, researchers propose decoupling factual knowledge from the core transformer weights, similar to how pre-trained visual features are aligned with large language models. This paradigm shift suggests that knowledge can be treated as an external, pluggable component, enabling efficient updates through modality bridging rather than invasive weight modifications that risk catastrophic forgetting or representation shattering.

Building on this visual analogy, spatially aware feature modulation techniques have emerged to precisely localize and manipulate knowledge representations within the transformer architecture. Methods such as neuron-specific masking and dynamic sparse masking leverage fine-grained attribution to identify distinct "knowledge neurons" in feedforward networks and attention layers. These approaches construct dynamic masks that isolate specific functional units, allowing for targeted interventions where editing signals are fused with token representations via intra-editing attention flows, thereby preserving the geometric integrity of unrelated semantic subspaces.

Furthermore, the integration of spatial awareness facilitates the propagation of editing signals without disrupting the global representation geom-

etry. By calculating token attention weights based on transformed representations, these methods ensure that new key-value pairs reshape hidden states only along relevant dimensions. This spatially constrained modulation effectively confines interference, offering superior scalability and locality compared to traditional weight-update strategies. Consequently, the model maintains its pre-edit performance on unrelated tasks while successfully incorporating new factual associations through structurally aware feature adjustments.

3.4.2 Language Anisotropic Updates and Cross Lingual Portability

Language anisotropic updates address the inherent directional bias in transformer representations, where knowledge is often encoded within specific subspaces that vary across linguistic structures. Standard isotropic editing methods frequently fail to account for these geometric disparities, leading to inefficient parameter utilization and degraded performance when applying edits across different languages. By constraining updates to orthogonal subspaces or leveraging low-rank adaptations aligned with language-specific manifolds, recent approaches ensure that modifications target only the relevant semantic dimensions without disturbing the broader representational geometry essential for multilingual coherence.

Cross-lingual portability remains a critical challenge, as edits performed in a source language often do not generalize effectively to target languages due to misaligned internal knowledge representations. Advanced techniques mitigate this by identifying shared latent concepts across languages and propagating updates through these invariant features rather than surface-level token mappings. This involves decomposing model weights to isolate language-agnostic knowledge components, allowing a single edit operation to simultaneously refine factual accuracy across multiple linguistic contexts while preserving the model's original multilingual capabilities and syntactic understanding.

The integration of anisotropic constraints with cross-lingual transfer mechanisms enables more robust and scalable lifelong editing scenarios. Methods employing dynamic auxiliary fusion or multiplicative orthogonal transformations demonstrate superior locality and generalization, ensuring that sequential updates in one language do not induce catastrophic forgetting in others.

These strategies collectively enhance the portability of edited knowledge, facilitating the deployment of large language models in global applications where consistent factual reliability across diverse linguistic communities is paramount without requiring exhaustive retraining for each language pair.

3.4.3 Unified Frameworks for Audio Visual Textual Editing

Unified frameworks for audio-visual-textual editing aim to synchronize semantic updates across heterogeneous modalities while preserving the integrity of pre-trained representations. A primary technical challenge involves resolving the dimensional mismatch between acoustic encoders and linguistic decoders. Contemporary approaches address this by leveraging projection layers to map audio editing residuals directly into the text representation space, ensuring that semantic modifications remain grounded in updated acoustic features. This cross-modal alignment is critical for maintaining consistency when edits propagate through multi-layered transformer architectures.

Recent methodologies favor localized, single-layer editing strategies over broad multi-layer updates to mitigate the risk of catastrophic forgetting and edit collapse. Empirical evidence suggests that precise interventions within the audio pathway often outperform diffuse multimodal variants, highlighting the necessity of granular control over parameter updates. By integrating speech-aware causal tracing, these frameworks can identify specific modules responsible for knowledge storage, enabling a "locate-then-edit" paradigm that minimizes interference with unrelated capabilities while maximizing editing reliability across diverse domains.

Furthermore, advanced architectures incorporate dynamic auxiliary fusion mechanisms to manage inter-editing attention flows, effectively capturing interactions between distinct factual updates. These systems balance the strong generalization of external hyper-networks with the computational efficiency of internal weight modifications, facilitating robust lifelong editing scenarios. By dynamically allocating update magnitudes and reducing the L1-norm of editing layers, unified frameworks achieve scalable performance, allowing for thousands of sequential edits without significant degradation in model fidelity or cross-domain transferability.

4 Activation Steering and Neuron Level Interventions

4.1 Activation Vector Construction and Manipulation

4.1.1 Contrastive Mean Activation Differences for Behavior Control

Contrastive Mean Activation Differences (CMAD) establish a foundational mechanism for behavior control by identifying causal directions within the model's latent space. This approach computes steering vectors through the arithmetic difference between mean activations elicited by positive and negative behavioral prompts, effectively isolating dimensions responsible for specific traits. Theoretical analyses, such as Difference-in-Means, suggest that this contrastive operation captures causally relevant pathways, allowing for precise intervention without altering underlying model weights. By averaging activation discrepancies across diverse example pairs, the method mitigates high variance inherent in single-instance comparisons, yielding robust vectors capable of generalizing across varied contexts.

The efficacy of CMAD relies heavily on the specificity of neuron selection and the stability of the resulting projection. Empirical evidence indicates that interventions targeting layers with high indirect effect scores, particularly within final MLP components, maximize behavioral modification while preserving fluency. However, the magnitude of the steering coefficient presents a critical trade-off: increasing the scaling factor amplifies the desired behavioral shift but risks degrading performance on downstream tasks due to excessive perturbation of the task subspace. Conversely, conservative scaling maintains general capabilities but may fail to fully suppress unwanted patterns, necessitating careful hyperparameter optimization to balance intervention strength against model integrity.

Advanced iterations of this paradigm extend beyond simple mean subtraction to incorporate contrastive learning objectives that disentangle target knowledge from unrelated representations. By maximizing mutual information between subject representations and target attributes while minimizing interference from orthogonal features, these methods enhance the precision of activation steering. Such refinements address limitations in traditional approaches where broad acti-

vation shifts inadvertently affect semantically adjacent concepts. Consequently, contrastive mean difference techniques have evolved into a versatile framework for real-time behavior alignment, offering a lightweight alternative to full fine-tuning that supports dynamic, human-interpretable control over large language model outputs.

4.1.2 Multi Steering via Vector Averaging and Simultaneous Injection

Multi-steering via vector averaging addresses the challenge of simultaneously enforcing multiple behavioral constraints by synthesizing distinct steering directions into a unified intervention vector [12]. This approach typically computes the arithmetic mean of individual task-specific vectors, such as those derived from contrastive activation differences for truthfulness, safety, or style. By aggregating these directions, the method aims to project the model’s hidden states toward a composite target subspace that satisfies all desired attributes without requiring separate inference passes. Theoretical analysis suggests that when steering vectors are approximately orthogonal in the activation space, linear superposition preserves the efficacy of each individual constraint, allowing for modular control over diverse model behaviors.

Simultaneous injection extends this concept by applying the aggregated steering vector directly to the residual stream of specific transformer layers during the forward pass. Unlike sequential editing techniques that modify weights permanently, this inference-time strategy adds a scaled offset to the activations, defined as $\mathbf{h}' = \mathbf{h} + \lambda \sum \mathbf{v}_i$, where λ governs the intervention strength. This mechanism enables dynamic adjustment of multiple traits, such as increasing formality while reducing toxicity, by modulating the magnitude of the combined vector. The injection point is critical; intervening at middle layers often yields the most robust multi-attribute control, as these layers encode high-level semantic representations suitable for broad behavioral modulation.

However, the reliability of vector averaging diminishes when constituent steering directions are non-orthogonal, leading to destructive interference where optimizing one attribute inadvertently degrades another. This phenomenon, often termed "steering collapse," occurs when the linear combination pushes activations into regions of the latent space that violate the manifold of coherent language generation. To mitigate this, ad-

vanced variants employ weighted averaging based on task priority or utilize optimization routines to find a Pareto-optimal direction that minimizes conflict. Despite these challenges, vector averaging remains a computationally efficient paradigm for multi-dimensional control, offering a lightweight alternative to retraining or complex router-based architectures.

4.1.3 Input Dependent Scaling and Component Level Editing

Input-dependent scaling addresses the limitations of uniform parameter updates by dynamically adjusting edit magnitudes based on specific input contexts and layer characteristics. Empirical evidence suggests that middle transformer layers exhibit higher separability metrics and contribute more significantly to behavioral modification than boundary layers. Consequently, adaptive scaling functions are employed to concentrate modification strength in these high-impact regions while reducing intensity near input and output boundaries. This strategy ensures that the scaling relative to the representation space, rather than absolute vector magnitude, governs effectiveness, thereby preserving input-output stability for out-of-scope data while maximizing specificity for targeted edits.

Component-level editing further refines this process by shifting from global weight adjustments to precise manipulations of internal mechanisms, such as individual neurons or attention keys. Methods like Neuron-level Sequential Editing utilize weight rewinding to preserve original model references while modifying hidden states, effectively mitigating the impairment of general capabilities. Research indicates that modifying attention keys often yields superior performance compared to value updates, achieving robust specificity with fewer trainable parameters. By identifying and perturbing influential neuron paths or multi-modal neurons, these techniques facilitate fine-grained control that minimizes side effects and maintains the integrity of the pre-trained knowledge distribution.

The integration of input-dependent scaling with component-level localization creates a unified framework for efficient model steering that balances edit success with locality preservation. Unlike fine-tuning, which redefines the global behavior landscape, this approach enables reversible, context-aware adaptations directly in activation or parameter space. By distributing total change

non-uniformly across targeted layers and restricting updates to a small subset of critical components, the method significantly reduces localization effort and computational overhead. This paradigm supports scalable industrial applications where timely, resource-efficient editing of factual associations is required without compromising the model’s broader generative capabilities or necessitating full retraining.

4.2 Feature Decomposition and Sparse Representation

4.2.1 Dictionary Learning and Monosemantic Feature Extraction

Dictionary learning has emerged as a pivotal technique for disentangling the polysemantic nature of neural representations within transformer models. By training sparse autoencoders on activation spaces, researchers decompose dense neuron activations into a high-dimensional basis of sparse features, each ideally corresponding to a single, interpretable concept. This approach addresses the fundamental limitation of standard basis neurons, which often encode multiple unrelated semantic meanings simultaneously due to superposition. The resulting monosemantic features provide a more granular and linearly separable representation of the model’s internal knowledge state, facilitating precise intervention without the collateral damage associated with modifying entangled neuron weights.

The extraction of these monosemantic features relies on optimizing a reconstruction loss regularized by an L1 penalty to enforce sparsity across the learned dictionary. Unlike traditional dimensionality reduction methods that prioritize variance retention, dictionary learning explicitly seeks a representation where individual features activate only in the presence of specific semantic contexts. Recent scaling laws suggest that increasing the dictionary size relative to the model dimension significantly improves feature interpretability and coverage, allowing for the isolation of rare or complex concepts that remain obscured in the native activation space. This process effectively transforms the opaque internal dynamics of large language models into a set of human-understandable variables that can be directly manipulated.

Consequently, monosemantic feature extraction enables advanced steering and editing capabilities by targeting specific knowledge components

rather than broad neuronal ensembles. When applied to knowledge modification tasks, this method allows for the selective amplification or suppression of factual associations by adjusting the coefficients of identified features during inference or fine-tuning. The composability of these sparse features further supports multi-dimensional control, where distinct concepts can be combined or separated to achieve nuanced behavioral changes. This paradigm shift from neuron-level to feature-level manipulation represents a critical advancement in ensuring the reliability and specificity of model edits while preserving overall utility.

4.2.2 Sparse Autoencoders for Fine Grained Intervention

Sparse Autoencoders (SAEs) have emerged as a pivotal mechanism for achieving fine-grained intervention in large language models by disentangling superimposed features within high-dimensional activation spaces. Unlike dense editing methods that risk catastrophic forgetting by modifying broad parameter sets, SAEs decompose activations into sparse, interpretable features, allowing for precise targeting of specific behavioral circuits. This decomposition leverages principles from compressed sensing, where sparse signals representing distinct concepts are recovered from dense neural representations, enabling interventions that alter specific knowledge without destabilizing the model’s general capabilities.

The efficacy of SAE-based intervention relies heavily on the sparsity of the learned representation to preserve utility. Empirical evidence suggests that dense ℓ_2 regularization often leads to severe performance collapse, whereas ℓ_0 -constrained sparse edits maintain downstream task accuracy by isolating modifications to a minimal subset of neurons. By identifying and manipulating only the most relevant features—often corresponding to specific entities or facts within feed-forward or attention modules—these methods achieve a superior trade-off between editing efficacy and locality. This precision ensures that the inherent capabilities of the pre-trained model remain intact, mitigating the interference typically observed in full fine-tuning or low-rank adaptation approaches.

Furthermore, the integration of SAEs facilitates scalable mass editing by operating directly on the activation space rather than weight matrices. This

approach allows for simultaneous updates to multiple facts through linear combinations of sparse feature vectors, significantly reducing computational overhead compared to iterative weight optimization. Recent advancements demonstrate that such neuron-level localization not only matches the performance of memory-based editing techniques on single-instance tasks but also sustains robustness when handling larger batches of edits. Consequently, SAEs provide a theoretically grounded framework for steering model behavior, offering a path toward reliable and interpretable model alignment in complex, multimodal architectures.

4.2.3 Disentangled Subject Representations and Knowledge Separation

Recent advancements in knowledge editing have shifted focus from holistic parameter updates to the precise disentanglement of subject representations within transformer architectures [13]. Theoretical analyses suggest that factual knowledge is not uniformly distributed but rather localized in specific mid-layer neurons where subject embeddings act as keys to retrieve relational objects. By isolating these "knowledge neurons," researchers can identify distinct subspaces responsible for encoding specific entities, thereby separating the representation of a subject from the broader contextual noise. This separation is critical for ensuring that modifications to one fact do not inadvertently alter unrelated knowledge stored in overlapping parameter regions.

$$\text{Edit Succ.} = \frac{1}{N} \sum_{(x_k, y_k^*)} \mathbb{1} \left\{ \arg \max_y f_{\theta'}(y \mid x_k) = y_k^* \right\}$$

Figure 11: Chart from 'Balancing Knowledge Updates: Toward Unified Modular Editing in LLMs' (arXiv:2510.27400)

The mechanism for achieving this separation often involves decomposing the feed-forward network operations into independent input and output directions. Studies indicate that valid rank-one updates must align strictly with the steering direction in the output space while remaining invariant to the input projection, effectively decoupling the storage of new information from the retrieval mechanism. This architectural insight allows for the manipulation of specific subject-rejection or subject-association vectors without disturbing the general linguistic capabilities of the model. Consequently, the model maintains high specificity,

updating only the targeted triplet while preserving the integrity of other facts associated with the same subject or relation.

Furthermore, empirical evaluations demonstrate that disentangled representations significantly mitigate the problem of catastrophic forgetting during knowledge erasure or insertion. When subject representations are successfully separated, editing operations result in minimal perplexity changes for unaffected relations, confirming that the model's internal knowledge graph remains structurally sound [14]. Techniques leveraging this principle have shown superior performance in handling sensitive personal information and correcting hallucinations compared to traditional fine-tuning methods. Ultimately, the ability to isolate and manipulate discrete knowledge components marks a pivotal step toward developing more transparent, editable, and reliable large language models capable of dynamic knowledge management.

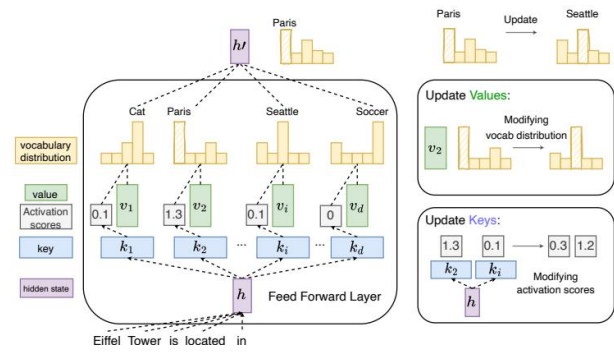


Figure 12: Chart from 'EMPIRICAL STUDY ON UPDATING KEY-VALUE MEMORIES IN TRANSFORMER FEED-FORWARD LAYERS' (arXiv:2402.12233)

4.3 Neuron Level Attribution and Targeted Modification

4.3.1 Integrated Gradients for Knowledge Neuron Identification

Integrated Gradients (IG) serve as a foundational attribution technique for localizing Knowledge Neurons within Transformer feed-forward networks, addressing the opacity of distributed knowledge representations. By computing the integral of gradients along a straight-line path from a baseline input to the actual input, IG assigns contribution scores to individual neurons relative to specific knowledge facts. This method effectively isolates the subset of parameters responsible for predicting target tokens, distinguishing genuine knowledge

carriers from correlated but causally irrelevant activations. The resulting attribution scores provide a granular map of where semantic information is encoded, enabling precise intervention without modifying the entire weight matrix.

To mitigate the computational burden of calculating continuous gradient integrals, practical implementations often employ Riemann sum approximations to discretize the integration path. In scenarios requiring batch editing of multiple knowledge facts, attribution scores are aggregated across samples to identify a unified Knowledge Neuronal Ensemble. This aggregation ensures that the selected neurons represent shared semantic structures rather than sample-specific noise. Furthermore, the method addresses the challenge of coupling in knowledge localization by summing neuron scores across a batch, thereby deriving a robust selection criterion that supports simultaneous updates to diverse knowledge sets while minimizing interference with unrelated model capabilities.

Once the critical neurons are identified via IG, the editing process restricts parameter updates solely to this ensemble, freezing the remainder of the network to preserve general linguistic competence. This targeted modification strategy contrasts with full fine-tuning, significantly reducing the risk of catastrophic forgetting and maintaining the stability of unmodified facts. By leveraging the high correlation between IG scores and semantic meaning, this approach ensures that edits are both efficient and effective, allowing for dynamic knowledge updates through constrained optimization on specific neuron subsets rather than broad architectural changes.

4.3.2 Key Value Memory Tuning versus Direct Value Modification

The conceptualization of Transformer feed-forward networks as key-value memory stores has precipitated a divergence in model editing strategies: tuning the retrieval mechanism versus modifying the stored content directly. Key-value memory tuning primarily adjusts the projection matrices responsible for generating keys, thereby altering the inner product dynamics between hidden states and memory slots. This approach effectively reroutes information flow by changing which neurons are activated for specific inputs without fundamentally rewriting the semantic representations encoded within the value vectors. Empirical evidence suggests that modifying

keys often yields superior locality and fluency compared to other layer edits, as it leverages the model’s existing distributed knowledge rather than imposing external constraints.

In contrast, direct value modification involves the explicit optimization of value projection weights or the injection of new factual vectors into specific neuron outputs. While this method allows for precise insertion of targeted knowledge, it frequently necessitates rigorous regularization to prevent catastrophic forgetting and distributional shifts. Direct updates require accurate optimization trajectories to ensure that the new values align with the pre-existing latent space; failure to do so can result in significant model degradation, particularly during sequential editing rounds. Consequently, direct modification often incurs higher computational costs and risks overfitting if the number of updated parameters is not strictly constrained to a minimal, influential subset.

Comparative analyses indicate that key-based tuning frequently outperforms direct value updates despite involving fewer trainable parameters. The superiority of key tuning stems from its ability to manipulate the retrieval probability distribution with minimal interference to the model’s core semantic storage. By focusing on the gating mechanisms, this strategy mitigates the impairment of general capabilities often observed when value vectors are forcibly overwritten. Furthermore, key tuning demonstrates greater robustness in multi-edit scenarios, as it avoids the cumulative parameter shift that plagues direct value modification, thereby preserving the integrity of the original model weights while achieving efficient, post-hoc knowledge integration.

4.3.3 Influential Path Editing for Cross Modal Unlearning

Influential path editing for cross-modal unlearning addresses the critical challenge of removing specific multimodal associations without degrading general model fidelity. Traditional locate-then-edit strategies often suffer from inaccurate localization, where causal tracing identifies broad layer associations rather than precise neuronal ensembles, leading to unintended retention of irrelevant knowledge. By shifting focus from static parameter modification to dynamic path intervention, these methods aim to isolate and sever the specific computational trajectories responsible for encoding unwanted cross-modal facts, thereby enhancing the

specificity of the unlearning process.

Advanced techniques leverage gradient attribution scores to identify influential neuron paths across modalities, ensuring that only parameters directly contributing to the targeted knowledge are updated. Unlike approaches that modify entire feed-forward network matrices via least-squares optimization, path-aware editing computes gradients specifically for the identified neuronal ensemble. This facilitates a collaborative update mechanism where error backpropagation is constrained to the influential path, preserving the integrity of unrelated representations and maintaining dynamic layer-wise interactions essential for coherent multimodal reasoning.

The efficacy of this approach lies in its ability to balance selective behavioral alteration with the preservation of global model capabilities. By enforcing constraints that keep disentangled, unrelated representations invariant during the editing phase, these methods mitigate the risk of catastrophic forgetting often observed in coarse-grained fine-tuning. Consequently, influential path editing provides a robust framework for machine unlearning in multimodal large language models, effectively eliminating specific concepts while sustaining performance on downstream tasks and ensuring consistency across diverse input modalities.

4.4 Expert Routing and Mixture of Experts Control

4.4.1 Router Function Adaptation for Behavior Suppression

Router function adaptation serves as a precise mechanism for behavior suppression in Mixture-of-Experts architectures by directly manipulating the gating logic that dictates expert selection. Rather than altering the internal weights of specific experts, this approach intervenes in the router’s output distribution to prevent the activation of neurons or experts correlated with undesired behaviors, such as refusal patterns or harmful outputs. By identifying and suppressing the top-k experts most relevant to specific negative traits, models can achieve significant behavioral shifts, such as a marked reduction in refusal rates, while maintaining performance on general benchmarks.

The technical implementation often involves modifying the softmax probabilities or zeroing out the routing weights assigned to targeted experts during the forward pass. This method ensures

that alignment-specific experts, which might otherwise be selected for certain tokens, are effectively bypassed without requiring full model retraining. Unlike activation steering which operates in hidden space, router adaptation modifies the discrete selection process itself, offering a localized control strategy. This is particularly effective when the undesirable behavior is sparse and attributable to a small fraction of the total expert population, allowing for high-granularity intervention with minimal computational overhead.

However, this technique faces challenges regarding the stability of expert selection and the potential for unintended side effects on unrelated tasks. Simply zeroing out router weights can disrupt the collaborative balance of the ensemble, potentially degrading the model’s ability to handle complex queries that require diverse expert contributions. Consequently, advanced variations employ scaling factors or dynamic constraints to modulate the extent of suppression rather than applying binary masks. This ensures that while specific behavioral modes are dampened, the overall functional fidelity of the network is preserved, striking a critical balance between selective behavioral alteration and the maintenance of general model capabilities.

4.4.2 Fixed Routing with Class Conditional Expert Alignment

Fixed routing with class conditional expert alignment establishes a deterministic mapping between input semantic categories and specific expert sub-networks within a Mixture of Experts architecture. Unlike dynamic gating mechanisms that rely on learned softmax probabilities to distribute tokens, this approach enforces a rigid topology where predefined classes activate designated experts. This structural constraint ensures that distinct knowledge domains are processed by isolated parameter sets, effectively mitigating the risk of expert collapse where a single dominant expert handles diverse inputs. By decoupling the routing logic from gradient-based optimization during inference, the model achieves consistent activation patterns that facilitate precise interpretability and targeted intervention.

The alignment mechanism operates by conditioning the expert selection process on explicit class labels or derived semantic clusters rather than latent feature similarities alone. During the forward pass, the router function is bypassed or

overridden to direct tokens belonging to a specific class c exclusively to a subset of experts E_c . This segregation allows for the preservation of specialized knowledge within dedicated sub-modules, preventing catastrophic interference when updating facts or adapting to new domains. Theoretical analysis suggests that fixing the routing matrix reduces the variance in expert utilization, thereby stabilizing the learning dynamics and ensuring that modifications to one expert do not inadvertently degrade the performance of unrelated tasks handled by other experts.

Implementing this strategy requires careful initialization to balance the load across the available expert pool while maintaining strict class separation. Although fixed routing sacrifices the flexibility of adaptive computation found in soft-gating methods, it offers significant advantages in safety-critical applications where predictable behavior is paramount. The deterministic nature of the pathway enables rigorous verification of information flow, ensuring that sensitive or aligned behaviors are confined to specific, auditable components. Furthermore, this modularity supports efficient parameter editing, as updates can be localized to class-specific experts without necessitating global retraining or complex regularization constraints to preserve unrelated capabilities.

4.4.3 Dynamic Expert Weighting for Alignment and Reasoning

Dynamic expert weighting mechanisms have emerged as a pivotal strategy for harmonizing alignment constraints with complex reasoning capabilities in Mixture-of-Experts architectures. Unlike static routing protocols, these approaches compute layer-wise interaction scores to identify specific neuronal ensembles critical for representing target knowledge. By leveraging gradient attribution and loss landscapes, the system facilitates dynamic information transfer across layers, ensuring that parameter updates are collaborative rather than isolated. This fine-grained localization allows the model to distinguish between parameters essential for factual retention and those required for logical deduction, thereby mitigating the risk of catastrophic forgetting during alignment procedures.

The core of this methodology involves manipulating router functions to modulate expert activation based on causal effects on output generation. During the editing or alignment process,

gradients are backpropagated exclusively through the identified knowledge neuronal ensemble, enabling precise weight adjustments without disrupting the broader parameter space. Techniques often employ load-balancing losses to prevent expert collapse, ensuring that multiple experts contribute to the weighted sum for complex reasoning tasks. This dynamic re-weighting allows the model to seamlessly adapt to new factual associations or safety guidelines by steering hidden states in crucial layers, effectively decoupling the storage of static knowledge from the execution of dynamic reasoning processes.

Empirical evaluations indicate that dynamic expert weighting significantly enhances both reliability and locality compared to full fine-tuning or static adapter methods. By isolating alignment-specific experts, these systems can suppress undesirable behaviors or inject new constraints while preserving general utility and reasoning depth. The approach demonstrates robustness in handling logical generalizations and maintaining relation specificity, as the modular control ensures that modifications to one knowledge domain do not inadvertently degrade performance in unrelated reasoning tasks. Consequently, this paradigm offers a scalable framework for maintaining up-to-date, safe, and logically coherent large language models in rapidly evolving informational environments.

5 Future Directions

Despite significant advancements in continual learning and model editing for foundation models, several critical limitations persist that hinder their deployment in truly dynamic environments. Current methodologies often struggle to maintain an optimal balance between stability and plasticity when faced with rapid, non-stationary data distributions, frequently resulting in subtle yet cumulative performance degradation over extended task sequences. While orthogonal constraints and dynamic merging offer promising avenues for isolating task-specific knowledge, they often incur substantial computational overhead during the inference phase due to the complexity of managing multiple adapter modules. Furthermore, existing unlearning techniques lack the precision required to remove specific harmful information or privacy-sensitive data without inadvertently degrading the model’s general utility or inducing collateral forgetting of related, benign concepts. The reliance

on frozen backbones, although efficient, restricts the model’s ability to fundamentally evolve its feature representations, potentially capping its adaptability in domains that diverge significantly from the original pre-training distribution.

Future research must prioritize the development of unified, scalable frameworks that seamlessly integrate acquisition, retention, and removal mechanisms into a single cohesive lifecycle management system. A promising direction involves the creation of self-regulating architectures capable of autonomously detecting distribution shifts and dynamically allocating resources between expanding capacity for new tasks and compressing or pruning obsolete knowledge. Investigating neuro-symbolic approaches could provide the structural inductive biases necessary to enhance interpretability and control during model editing, ensuring that knowledge modifications adhere to logical constraints and safety guidelines. Additionally, there is an urgent need to explore theoretical bounds on the capacity of low-rank adapters and sparse expert networks to determine the fundamental limits of parameter-efficient continual learning. Developing robust metrics for evaluating the long-term integrity of edited models, beyond simple accuracy benchmarks, will be essential to quantify the extent of latent knowledge corruption and ensure reliable operation in high-stakes applications.

The successful realization of these future directions holds the potential to fundamentally transform the deployment paradigm of artificial intelligence, shifting from static, periodically retrained systems to truly lifelong learners. By enabling foundation models to adapt continuously while rigorously maintaining safety and privacy standards, the proposed research will facilitate the creation of resilient AI systems capable of operating autonomously in complex, real-world scenarios such as personalized healthcare, autonomous navigation, and dynamic financial modeling. This evolution will not only reduce the massive computational and environmental costs associated with full model retraining but also democratize access to advanced AI by allowing smaller organizations to maintain up-to-date models through efficient, targeted updates. Ultimately, bridging the gap between continual learning and precise model editing will establish a new standard for responsible AI, ensuring that intelligent systems remain aligned with human values and operational requirements throughout their entire operational

lifespan.

6 Conclusion

This survey has systematically synthesized the converging landscapes of continual learning and model editing within the era of foundation models, highlighting critical strategies for maintaining model relevance in dynamic environments. We examined sophisticated mechanisms designed to mitigate catastrophic forgetting, ranging from orthogonal subspace constraints that enforce geometric separation of task-specific updates to replay-free knowledge distillation techniques that preserve prior distributions without historical data storage. Furthermore, the discussion detailed parameter-efficient adaptation architectures, emphasizing low-rank decomposition and modular adapter compositions that drastically reduce optimizable parameters while retaining generalization power. Beyond knowledge acquisition, we addressed the imperative of unlearning and selective knowledge removal, demonstrating how gradient-based forgetting and representation nullification enable precise excision of harmful or obsolete information. By integrating these acquisition and removal paradigms, the survey establishes a comprehensive understanding of how foundation models can evolve through precise architectural interventions rather than prohibitive full retraining.

The significance of this work lies in its establishment of a unified framework that bridges the traditionally isolated fields of continual learning and model editing. As the scale of neural architectures grows, the computational and economic costs of static training paradigms become increasingly unsustainable, making the ability to adapt sequentially and edit selectively a prerequisite for responsible and scalable AI deployment. By providing a systematic taxonomy of existing methods categorized by their mathematical principles and architectural modifications, this survey offers a critical lens through which researchers can evaluate trade-offs in memory efficiency, computational overhead, and performance retention. It elucidates the delicate balance between stability and plasticity, offering actionable insights for developing systems that do not merely accumulate knowledge but also curate it. This holistic perspective is essential for advancing the lifecycle management of foundation models, ensuring they remain robust, ethically aligned, and capable of operating effectively

in non-stationary real-world scenarios.

Looking forward, the integration of continual learning and model editing must transition from theoretical exploration to standardized practice in next-generation AI systems. Future research should prioritize the development of automated, self-correcting frameworks that can dynamically decide when to learn new tasks, merge existing knowledge, or excise harmful behaviors without human intervention. There is an urgent need for rigorous benchmarking standards that evaluate models not only on static accuracy but also on their long-term adaptability, forgetting rates, and editing precision across diverse domains. Practitioners and researchers are called to collaborate on creating open, modular architectures that facilitate seamless updates and safe deployment in sensitive applications such as healthcare and finance. Ultimately, realizing the full potential of lifelong learning in foundation models requires a concerted effort to refine these adaptive mechanisms, ensuring that artificial intelligence evolves as a resilient, trustworthy, and continuously improving partner in solving complex global challenges.

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